MCA SOFTWARE ENGINEERING - TUTORIAL 6

Sequence Diagrams: Numerical & Problem-Solving Practice

These are sequence-based problem-solving questions. Sequence diagrams do not normally use mathematical numericals; these questions use counting, ordering and repetition to build logical understanding.

Q1. Count the interaction [Easy]

A login process has these messages:
1. Student -> Login System: enterLogin()
2. Login System -> Database: checkUser()
3. Database -> Login System: userValid
4. Login System -> Student: loginSuccess

Find: (a) number of lifelines, (b) number of messages, (c) first message, (d) last message.

Answer:

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Q2. Repeated book issue [Easy-Medium]

For issuing books, checkMember() is performed once. For every book, the system performs checkBook(), createLoan(), and showDueDate().
A student issues 4 books. Calculate the total number of messages. Also identify which part can be shown using a loop fragment.

Answer:

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Q3. Alternative login paths [Medium]

A login request always sends enterCredentials() and validateUser(). If credentials are valid, the system sends openDashboard(). If invalid, it sends showError().
During 10 login attempts, 7 are valid and 3 are invalid.
(a) How many total messages are sent across all 10 attempts?
(b) Which UML combined fragment represents the two possible paths?
(c) Write the two guard conditions.

Answer:

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Q4. Arrange the messages [Medium]

The following Return Book messages are mixed:
A. calculateFine()
B. returnBook()
C. updateBookStatus()
D. checkDueDate()
E. displayFine()

Arrange them in a logical order and then draw a sequence diagram using Member, Librarian, Library System and Database.

Answer:

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Q5. Library issue-book challenge [Medium]

An Issue Book scenario uses:
requestIssue(), enterDetails(), validateMember(), checkAvailability(), createLoan(), showDueDate().
If the book is unavailable, createLoan() and showDueDate() are replaced by showNotAvailable().
(a) How many messages occur in the available path?
(b) How many messages occur in the unavailable path?
(c) Which fragment should be used to show these alternatives?
(d) Draw the sequence diagram.

Answer:

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